

State of Digital Health in 32 Countries of the WHO African Region: A Multi-Country Maturity Assessment with Short-Term Longitudinal Comparisons Across Leadership, Policy, and Health Systems Domains (2023 - 2025)

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Background: Evidence on digital health maturity in Africa remains limited. We characterized maturity across countries in the World Health Organization African Region between 2023 and 2025 to generate a regional baseline for policies and investments.

Methods: We analyzed the Global Digital Health Monitor data, which scored seven domains of the digital health enabling environment on a 5-phase scale. All assessments from the 2023, 2024, and 2025 cycles were extracted in April 2026, yielding 50 country-year observations from 32 of the 47 member states of the World African Region, with 17 of these 32 countries reassessed in at least one subsequent cycle. Assessment coverage declined over time (29, 17, and four countries in 2023, 2024, and 2025, respectively); therefore, the 2025 comparison rests on a small subset. The data are self-reported by the country focal points and published only after government approval. We used descriptive statistics (means with medians and interquartile ranges for the ordinal phase scale), choropleth mapping, short-term descriptive change analysis, Spearman correlation (reported with sample size and *p*-values), hierarchical clustering with internal validation (silhouette score, gap statistic, and linkage sensitivity), and within-country gap analysis.

Results: We analyzed 32 countries and 50 country-year observations. Most countries were in Phase 3 (established; 21 of 32, 66%), with none in phases 1 or 5; 5 were at Phase 4, and 6 were at Phase 2. Leadership and governance were consistently the strongest (mean phase 3.00 in 2023 and 3.75 in 2025), and the workforce was the weakest. Of the 17 reassessed countries, three improved and none declined; one country (Kenya) advanced two phases in one year. Standards and interoperability were strongly correlated with services and applications (Spearman $\rho = 0.70$, $n = 30$, $p < 0.001$). Clustering yielded three archetypes: infrastructure-oriented, governance-led, and early stage. A total of 26 of the 32 countries had gaps of two or more phases between the strongest and weakest domains, typically strong governance with a weak workforce.

Conclusion: Among the assessed countries in the World Health Organization African Region, digital health rests on established governance but is constrained by a weak workforce and uneven interoperability. Targeted investment in the workforce, shared data architectures, and broader maturity monitoring are the most consequential levers for progress in this field. The findings should not be extrapolated to non-assessed Member States or to the wider African continent, which extends beyond the World Health Organization's African Region.

Keywords: digital health, Africa, World Health Organization African Region, health systems, maturity assessment, workforce, governance, interoperability

Introduction

Digital health technologies, including electronic health records (EHRs), mobile and telehealth services, data-driven decision support, geospatial surveillance, and artificial intelligence (AI) applied to clinical and public health practice, have moved from the periphery of health system reform to its center.¹ Global institutions now frame digital transformation as a structural determinant of health system performance, quality of care, and equitable access, and the World Health Organization (WHO) Global Strategy on Digital Health explicitly positions digital solutions as an enabler of primary health care, universal health coverage, and progress towards the Sustainable Development Goals.² The Global Strategy emphasizes that sustainable digital transformation requires coordinated action across governance, workforce, standards, infrastructure, and service delivery, rather than the isolated deployment of technologies.³ In this study, we use the term digital health maturity to denote the cumulative readiness of a country's digital health enabling environment (its governance, strategy, legislation, workforce, standards, infrastructure, and services), measured against a stage-based rubric, in line with how the construct has been operationalized in recent maturity-model syntheses.⁴ This usage is distinct from related but narrower constructs such as digital health adoption (whether tools are in use), readiness (capacity to introduce them), implementation (how they are deployed), and outcomes (their measurable impact on services and population health).⁵

The WHO African Region illustrates both the promise and the fragility of this agenda. Its 47 Member States are home to approximately 1.2 billion people, with a median age below 20 years and life expectancy at birth that reached about 64 years in 2019, rising faster than in any other WHO region but still notably below high-income country averages this demographic and disease landscape sustains a continuing imperative for equitable scale-up of primary health care.^{6,7} The Region bears a substantial share of the global burden of communicable and non-communicable diseases.⁸ It also faces persistent shortages of health workers and depends on fragmented and often paper-based information systems that constrain the accuracy and timeliness of decision-making.⁹ Simultaneously, mobile penetration, digital public infrastructure, and regional cooperation through the African Union, Africa Centres for Disease Control and Prevention (Africa CDC), and regional economic communities have created an unprecedented platform for scaling digital health solutions aligned with continental priorities on pandemic preparedness, primary health care, and universal health coverage.¹⁰ Ministries of Health across the Region are actively formulating or revising national digital health strategies, establishing governance structures, and investing in integrated EHRs, health data exchange, and digital workforce development.¹¹

The coronavirus disease 2019 (COVID-19) pandemic was pivotal for digital health in Africa. Response activities between 2020 and 2022 accelerated the deployment of telehealth, electronic surveillance, laboratory information systems, and digital credentials, while exposing persistent weaknesses in governance, interoperability, workforce readiness, cybersecurity, and equitable access.¹² The pandemic experience crystallized the consensus that digital health should be treated as a public good anchored in robust institutional foundations rather than a collection of time-limited projects.¹³ Development partners fund most national digital health activities in the African Region, and the WHO Regional Office for Africa, based in Brazzaville, Republic of Congo, has adopted a framework to guide Member States and partners in strengthening governance, workforce, standards, and sustainable financing for digital health.^{14,15} In parallel, continental institutions, including Africa CDC, have positioned digital transformation and health information exchange as strategic priorities for strengthening health systems across the Region.¹⁶

Despite this momentum, comparable multi-country evidence on the state of digital health in Africa is limited. Prior assessments have often focused on single-country case studies, specific technologies such as electronic medical records or mobile health, or narrow dimensions of digital readiness.¹⁷ The absence of a harmonized measurement framework has made it difficult to compare maturity across countries, distinguish genuine progress from strategy announcements, and identify where investment is most needed.⁹ In response to this gap, the Global Digital Health Monitor (GDHM), which evolved from the Global Digital Health Index first released in 2016 and was redesigned in 2022 to align with the Global Strategy on Digital Health, offers a standardized 5-phase rubric spanning seven domains of the digital health enabling environment: leadership and governance, strategy and investment, legislation and policy, workforce, standards and interoperability, infrastructure, and services and applications.¹⁸ Country partners and ministries of health submit and validate data through an annual survey cycle, and the resulting indicator-level and overall country phase scores support

structured cross-country comparisons. The GDHM offers the largest harmonized, openly accessible measurement instrument currently available for cross-country digital health maturity, but it also has known constraints: data are self-reported by country partners, only government-approved submissions are published, participation is voluntary, indicator anchors are necessarily stylized, and the platform measures the enabling environment rather than realized service delivery or health outcomes.¹⁸ These features inform both the strengths and limitations of our analysis.

To our knowledge, no peer-reviewed analysis has systematically characterized the state of digital health maturity across countries in the WHO African Region using the most recent GDHM cycles, compared the relative strengths and weaknesses of the seven domains, mapped the geographic distribution of maturity, examined short-term descriptive change among countries reassessed over successive cycles, or identified country archetypes that could inform tailored technical assistance. We note that the WHO African Region comprises 47 Member States and excludes the seven African countries classified in the WHO Eastern Mediterranean Region (Djibouti, Egypt, Libya, Morocco, Somalia, Sudan, and Tunisia); therefore, our findings apply to Member States of the WHO African Region rather than to the entire African continent.^{19,20} Such a baseline is needed to support ministries of health, regional bodies, donors, and implementing partners in setting priorities, benchmarking progress, and directing finite resources to the domains where they will yield the greatest return on investment, particularly as the region moves towards the Global Strategy 2030 horizon and integrates AI and large-scale data systems into national health programmes.^{21,22} This study aims to describe and interpret the state of digital health maturity across countries of the WHO African Region between 2023 and 2025, using all available GDHM data to characterize the regional profile, map the geographic distribution of maturity, analyze short-term change among repeat-assessed countries, examine domain correlations and identify country archetypes through hierarchical clustering, and quantify within-country maturity gaps, in order to generate a baseline that can inform digital health policy, investment, and future measurement across the WHO African Region.

Methods

Study Design, Setting, and Reporting

We conducted a cross-sectional secondary analysis of publicly available aggregated country-level data with a short-term longitudinal descriptive component examining digital health maturity between 2023 and 2025. This study was reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement for cross-sectional studies, and a completed STROBE checklist is provided in [Supplementary S1](#). The setting was the 47 Member States of the WHO African Region, the regional grouping used by the GDHM to report maturity data for Africa, and the unit of analysis was the country-year. Within-country changes were descriptively examined for countries assessed in multiple GDHM cycles.

Data Source and Framework

Data were obtained from the GDHM, an interactive web-based platform maintained by HealthEnabled in partnership with the WHO and global health stakeholders.¹⁸ Country-level data for the Member States of the WHO African Region across the 2023, 2024, and 2025 assessment cycles were extracted in April 2026. The platform assesses digital health maturity through a structured framework of seven indicator categories containing 23 indicators (31 individual items including sub-indicators), with each indicator scored on a 5-phase continuum ranging from a nascent state with no formal structure or capacity to an optimized state with full institutionalization and comprehensive coverage.

The seven indicator categories are leadership and governance, strategy and investment, legislation and policy, workforce, standards and interoperability, infrastructure, and services and applications. A complete list of the 23 indicators (31 items with sub-indicators), their assessment questions, and the descriptive anchors for each of the five maturity phases is provided in [Supplementary S2](#). Country partners and Ministry of Health representatives submit data through an annual survey, and only government-approved submissions are published in the database.

Study Population and Variables

We included all WHO African Region Member States that had completed at least one GDHM assessment between 2023 and 2025, capturing 32 of the 47 member states, together with all available assessment years per country, yielding an unbalanced panel of 50 country-year observations. The remaining 15 Member States either did not participate in any of the three cycles or had not received the required government approval for publication during the study window; non-participation reflects the voluntary, government-validated design of the platform rather than an explicit exclusion criterion applied by the investigators.¹⁸ For each country-year observation, we extracted the country name, International Organization for Standardization (ISO) 3166 alpha-3 code, assessment year, overall country phase, seven category-level phase values, and seven category-level composite scores. For analyses that required a single observation per country (regional profile, ranking, correlation, clustering, and gap analysis), we used each country's latest available assessment year, defined as the most recent published cycle on the platform as of April 2026.

Data Cleaning, Bias, and Missing Data

Three data points coded as pending government approval (Niger infrastructure 2023 and Togo strategy and investment for 2023 and 2024) were recoded as missing and excluded from the affected analyses, with no imputation performed. The extracted data were validated programmatically and visually against the live platform to confirm their concordance. Potential sources of bias documented for self-reported, government-approved digital health monitoring instruments, including the GDHM itself and broader reviews of digital health implementation in sub-Saharan Africa, were considered in our interpretation.^{9,18} These include social desirability and selection biases from country self-reports, suppression of less favorable results through the government-approval gate, and over-representation of countries with active digital health partnerships or monitoring capacity through voluntary participation.

Statistical Analysis

All analyses were descriptive and exploratory and were conducted using Python 3.11. Because phase values are ordinal on a 1-to-5 scale, we report medians and interquartile ranges (IQRs) alongside means in all category-level summaries; means are retained for comparability with previous digital health maturity reports. For the regional profile, we computed the frequency distributions of overall country phases using each country's latest assessment, category-level summary statistics stratified by year, and country ranking by overall phase and composite score using each country's latest assessment. For the spatial analysis, we generated choropleth maps of the overall country phase using a 5-class sequential color scale, with seven additional maps for each indicator category and a 3-panel temporal comparison across the 2023, 2024, and 2025 cycles.

Non-assessed Member States of the WHO African Region are displayed with grey hatching, and African countries outside the WHO African Region are visually distinguished and excluded from the analytical sample. For countries with repeated assessments, we computed the phase difference between the earliest and latest assessments for the overall phase and for each category, visualized country trajectories with a slope graph, and displayed category-level changes using a heatmap. Given the unbalanced panel and the short observation window of 2 to 3 years, no inferential trend modelling was applied; all longitudinal analyses were treated as short-term descriptive changes rather than as a test of robust longitudinal progression.

To examine domain interdependencies, we calculated Spearman rank correlations among the seven category composite scores using the latest assessment per country and presented the results as a lower triangular heatmap. Hierarchical agglomerative clustering with Ward's linkage on z-standardized category scores was used to identify the country archetypes. The number of clusters was selected after triangulating three sources of evidence: visual inspection of the dendrogram, silhouette analysis across 2-, 3-, and 4-cluster solutions, and the gap statistic computed against uniform reference distributions using 100 bootstrap samples.^{23,24} We additionally repeated the clustering with single, complete, and average linkage for sensitivity analysis. A gap analysis summarized the phase distribution across categories and computed within-country gaps as the difference between each country's maximum and minimum category phases. The sample size and *p*-values for each pairwise correlation, Kendall tau-b cross-checks, cluster-validation outputs (silhouette and linkage sensitivity, gap statistic), cluster summary statistics, country-by-year coverage matrix, and year-stratified category phase distributions with medians and interquartile ranges are reported in [Supplementary S3](#).

Results

Regional Profile of Digital Health Maturity

The analytic sample covered 32 Member States of the WHO African Region and 50 country-year observations, of which 15 countries contributed a single assessment, 16 contributed two assessments, and one country (Mali) contributed three assessments. Assessment coverage declined over time: 29, 17, and 4 countries were assessed in 2023, 2024, and 4 in 2025 (Table 1). Based on each country's latest available assessment, most countries were classified as Phase 3,

Table 1 Summary Statistics for the Overall Maturity Phase and Each of the Seven Indicator Categories Across Member States of the WHO African Region by Assessment Year, 2023 to 2025

Year	Category	N	Mean Phase	Median Phase	Interquartile Range	Range	Mean Score
2023	Leadership & Governance	29	3.0	3.0	2.0–3.0	2-5	2.66
	Strategy & Investment	28	2.82	3.0	2.0–3.0	1-5	2.68
	Legislation, Policy & Compliance	29	2.83	3.0	2.0–3.0	1-4	2.51
	Workforce	29	2.0	2.0	1.0–2.0	1-5	1.79
	Standards & Interoperability	29	2.38	2.0	2.0–3.0	1-5	2.26
	Infrastructure	28	2.57	3.0	2.0–3.0	1-4	2.41
	Services & Applications	29	2.55	2.0	2.0–3.0	1-4	2.2
	Overall	29	2.83	3.0	2.0–3.0	2-4	n/a
2024	Leadership & Governance	17	3.29	4.0	2.0–4.0	2-5	3.01
	Strategy & Investment	16	3.06	3.0	2.75–4.0	2-4	2.94
	Legislation, Policy & Compliance	17	2.94	3.0	3.0–3.0	1-4	2.62
	Workforce	17	2.29	2.0	2.0–3.0	1-4	2.04
	Standards & Interoperability	17	2.71	2.0	2.0–3.0	1-5	2.53
	Infrastructure	17	2.47	2.0	2.0–3.0	1-4	2.35
	Services & Applications	17	2.82	3.0	2.0–4.0	1-4	2.47
	Overall	17	3.06	3.0	3.0–3.0	2-4	n/a
2025	Leadership & Governance	4	3.75	3.5	3.0–4.25	3-5	3.31
	Strategy & Investment	4	3.5	3.5	3.0–4.0	3-4	3.25
	Legislation, Policy & Compliance	4	3.0	3.0	2.75–3.25	2-4	2.5
	Workforce	4	3.5	3.5	2.75–4.25	2-5	3.0
	Standards & Interoperability	4	2.75	2.5	2.0–3.25	2-4	2.5
	Infrastructure	4	3.0	3.0	2.75–3.25	2-4	2.75
	Services & Applications	4	3.5	3.5	3.0–4.0	3-4	3.05
	Overall	4	3.25	3.0	3.0–3.25	3-4	n/a

Notes: Phase values range from 1 (nascent) to 5 (optimised). N is the number of countries contributing data to that domain in that year. Mean phase and median phase are the mean and median of the integer phase value; the interquartile range gives the 25th and 75th percentiles. Mean score is the mean of the continuous category composite score. Overall refers to the country-level overall maturity phase; "n/a" indicates that the mean composite score is not computed for the overall row because the composite is derived per category, not for the overall phase.

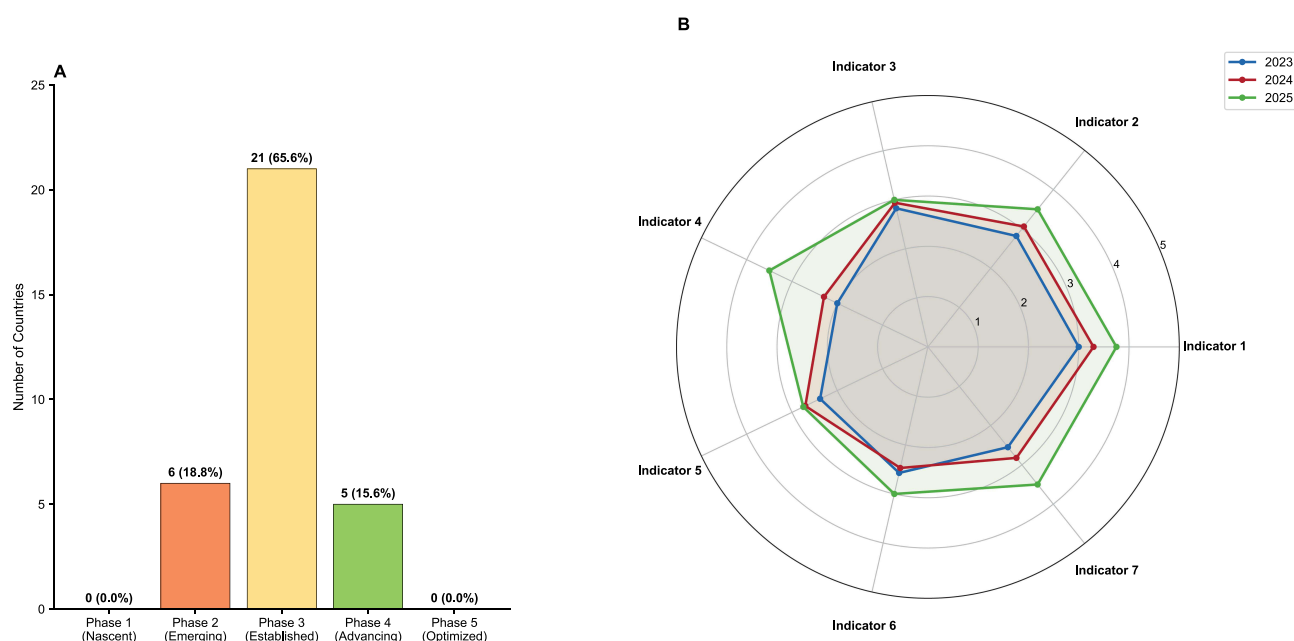


Figure 1 Distribution of overall digital health maturity phases and category phase profiles for Member States of the WHO African Region, 2023 to 2025. **(A)** displays the bar chart of countries by overall phase based on each country's latest assessment (32 countries; 0/6/21/5/0 across phases 1 to 5). **(B)** is a radar plot of the regional mean category phase by assessment year; the three year-specific series are based on different country samples (29 in 2023, 17 in 2024, 4 in 2025) and are therefore not directly comparable. Indicators 1 to 7 denote leadership and governance, strategy and investment, legislation and policy, workforce, standards and interoperability, infrastructure, and services and applications, respectively.

corresponding to an established level of digital health maturity, with 21 of the 32 countries at this level. A further six countries were in phase 2, described as emerging, and five were in phase 4, described as advancing; no country reached Phase 1 or Phase 5 for overall maturity (Figure 1).

Across the three assessment cycles, leadership and governance was consistently the strongest domain, with mean phase values of 3.00 in 2023, 3.29 in 2024, and 3.75 in 2025, followed by legislation and policy, and strategy and investment. Workforce was consistently the weakest domain, with mean phase values of 2.00 in 2023, 2.29 in 2024, and 3.50 in 2025; the higher 2025 value rests on only four countries. Services and applications occupied an intermediate position over the years (Table 1). Across the 21 domain-by-year cells, median phase values differed from the corresponding means by at most 0.71, and the qualitative ranking of domains was preserved when the medians replaced the means. The full medians and IQRs are reported in Table 1.

The country ranking by overall phase and composite score for each country's latest assessment placed Tanzania at the top with a composite score of 3.60, followed by Malawi, Guinea, Kenya, and Ethiopia, all in Phase 4. The lowest-ranked countries were Guinea-Bissau, Comoros, Togo, Gabon, and Mauritania, all in phase 2, with composite scores below 2.0 (Table 2).

Geographic Distribution of Maturity

Mapping the overall country phase by geographic location did not reveal a single dominant regional cluster, although several patterns were observed (Figure 2), as follows: the five phase-4 countries were distributed across East, West, and Southern Africa, whereas phase-2 countries were concentrated in West and Central Africa, including Comoros, Gabon, Guinea-Bissau, Mauritania, Nigeria, and Togo. Fifteen African region member states were not assessed during the study period and are represented by grey hatching on the maps. Countries on the African continent that lie outside the WHO African Region were excluded from this analysis.

Category-specific maps revealed variations between domains (Figure 3). Leadership and governance showed the widest distribution of higher phases, with several countries reaching Phase 4 or Phase 5. In contrast, workforce,

Table 2 Country Ranking by Overall Digital Health Maturity Phase and Composite Score Using Each Country's Latest Assessment

Rank	Country	Assessment Year	Overall Phase	Composite Score
1	Tanzania	2024	4	3.6
2	Malawi	2024	4	3.54
3	Guinea	2024	4	3.46
4	Kenya	2024	4	3.43
5	Ethiopia	2025	4	3.35
6	Mali	2025	3	3.01
7	Benin	2023	3	2.94
8	Mozambique	2023	3	2.87
9	Botswana	2025	3	2.86
10	Zambia	2024	3	2.74
11	Zimbabwe	2024	3	2.68
12	South Africa	2023	3	2.64
13	Liberia	2024	3	2.61
14	Ghana	2023	3	2.61
15	Sierra Leone	2024	3	2.59
16	Rwanda	2023	3	2.52
17	Madagascar	2024	3	2.5
18	Namibia	2023	3	2.46
19	Burkina Faso	2025	3	2.41
20	Eswatini	2024	3	2.35
21	Burundi	2024	3	2.28
22	Chad	2024	3	2.26
23	Democratic Republic of the Congo	2023	3	2.25
24	Niger	2023	3	2.19
25	Cameroon	2023	3	2.17
26	Cape Verde	2024	3	2.08
27	Nigeria	2023	2	2.01
28	Mauritania	2023	2	1.95
29	Gabon	2024	2	1.75
30	Togo	2024	2	1.48
31	Comoros	2024	2	1.44
32	Guinea-Bissau	2023	2	1.14

Notes: Ranking is based on each country's latest available Global Digital Health Monitor assessment, defined as the most recent published cycle on the platform as of April 2026. Composite score is the mean of the 7 category composite scores. Countries appear once in this table even if they were assessed in multiple cycles; see Table 3 for the subset of repeat-assessed countries.

Table 3 Phase Change and Big Movers Among Member States of the WHO African Region with Repeat Global Digital Health Monitor Assessments

Country	Earliest Year	Latest Year	Δ Overall	Δ Leadership & Governance	Δ Strategy & Investment	Δ Legislation & Policy	Δ Workforce	Δ Standards & Interoperability	Δ Infrastructure	Δ Services & Applications	Big Movers (2+ Phases)
Burundi	2023	2024	0	0	0	0	0	0	0	0	None
Cape Verde	2023	2024	0	0	0	0	0	-2	0	0	None
Chad	2023	2024	1	0	2	0	1	1	0	0	Strategy & investment (+2)
Comoros	2023	2024	0	0	0	0	0	0	0	0	None
Ethiopia	2023	2025	0	0	0	0	0	0	0	1	None
Gabon	2023	2024	0	1	0	0	0	0	0	0	None
Guinea	2023	2024	0	0	0	0	0	0	0	0	None
Kenya	2023	2024	2	2	0	2	1	3	1	2	Leadership & governance (+2); Legislation & policy (+2); Standards & interoperability (+3); Services & applications (+2)
Liberia	2023	2024	0	-1	0	1	1	1	0	2	Services & applications (+2)
Madagascar	2023	2024	0	1	1	0	1	0	0	0	None
Malawi	2023	2024	0	-1	0	0	0	0	0	-1	None
Mali	2023	2025	0	2	-1	1	1	0	0	0	Leadership & governance (+2)
Sierra Leone	2023	2024	1	2	1	0	1	0	1	0	Leadership & governance (+2)
Tanzania	2023	2024	0	0	0	0	0	0	0	0	None
Togo	2023	2024	0	0	n/a	0	0	0	0	0	None
Zambia	2023	2024	0	0	1	1	0	1	0	0	None
Zimbabwe	2023	2024	0	1	0	0	1	0	0	0	None

Notes: Δ (delta) denotes the change in phase between a country's earliest and latest assessment. Zero indicates no change, positive values denote improvement, and negative values denote a decrease. "Big movers" lists categories where a country improved by 2 or more phases; "none" indicates no category met the threshold.

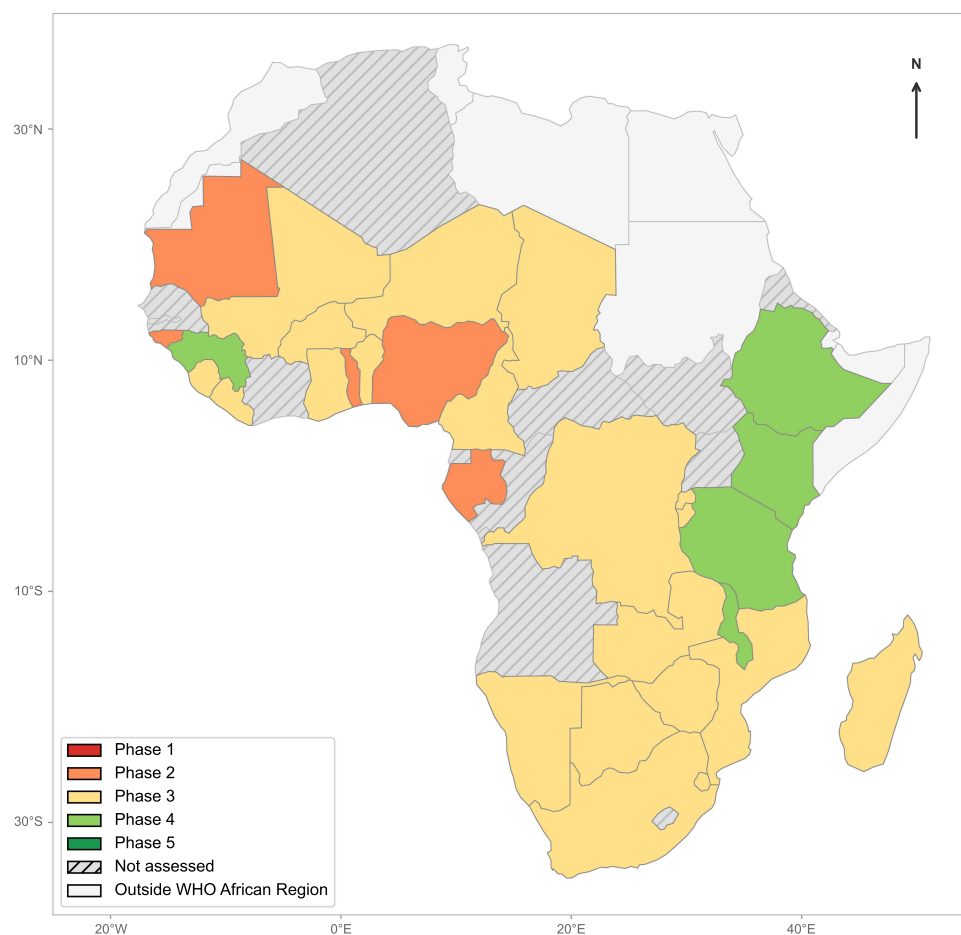


Figure 2 Choropleth map of overall digital health maturity phase for each Member State of the WHO African Region in its latest assessment year. Countries are shaded by phase (2 to 4); non-assessed Member States of the WHO African Region are shown with grey hatching, and African countries outside the WHO African Region are visually distinguished and excluded from the analytic sample.

standards, and interoperability maps were dominated by Phase 1 and Phase 2 values across most regions. A 3-panel temporal comparison indicated that assessment coverage declined over time, with 29, 17, and 4 countries assessed in 2023, 2024, and 2025, respectively (Figure 4).

Short-Term Change in Repeat-Assessed Countries

Among the 17 countries with repeat assessments, the mean change in the overall phase from the earliest to latest assessment was 0.24. Three countries improved in the overall phase, 14 showed no change, and none showed a decline. Kenya showed the largest overall improvement, advancing from Phase 2 to Phase 4 between 2023 and 2024, whereas Chad and Sierra Leone each improved from Phase 2 to Phase 3 (Figure 5).

Category-level changes were more frequent than the overall phase changes. Eight instances of two- or multi-phase improvements in a single category were identified across five countries. Kenya exhibited the most pronounced gains, with 2-phase improvements in leadership and governance, legislation and policy, and services and applications, and a 3-phase improvement in standards and interoperability. Two-phase improvements were also observed in Chad for strategy and investment, Liberia for services and applications, Mali for leadership and governance, and Sierra Leone for leadership and governance (Table 3).

Domain Correlations and Country Archetypes

Spearman rank correlation analysis of the 7 category composite scores, restricted to the 30 countries with complete data on all categories, revealed predominantly positive correlations among the domains (Figure 6).

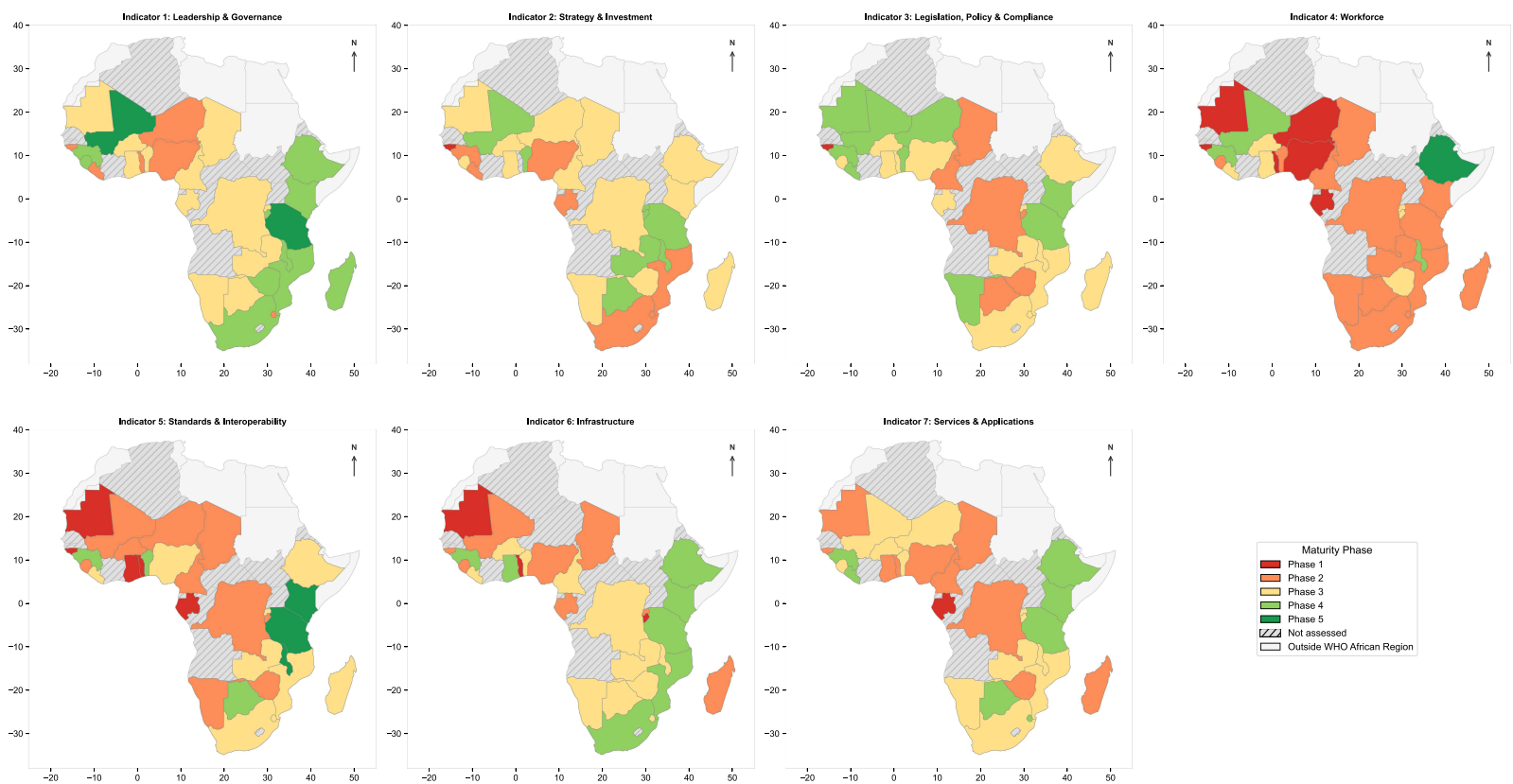


Figure 3 Category-specific choropleth maps of digital health maturity across the 7 GDHM domains (leadership and governance, strategy and investment, legislation and policy, workforce, standards and interoperability, infrastructure, and services and applications).

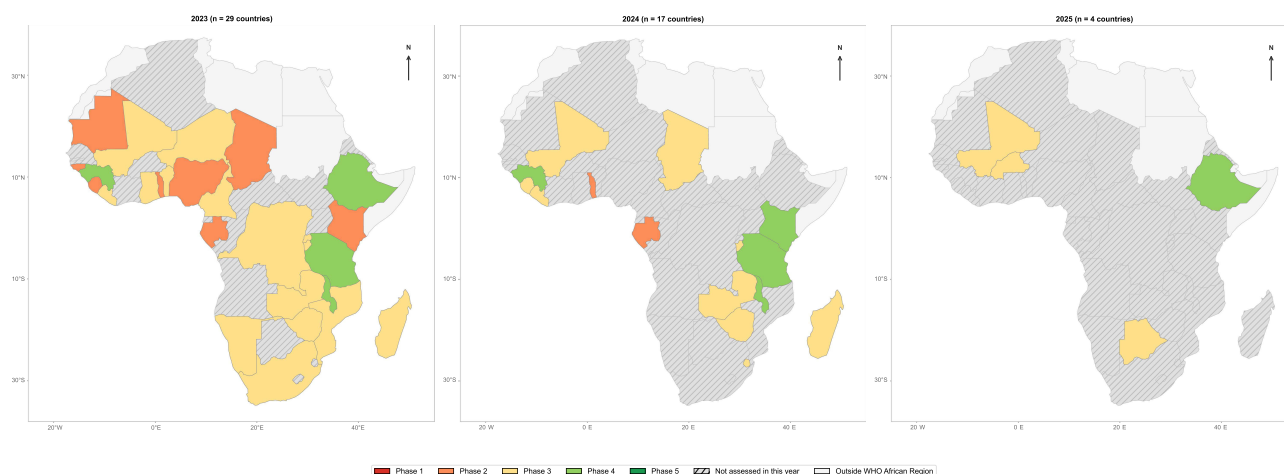


Figure 4 Temporal comparison of country-level overall maturity phases across the 2023, 2024, and 2025 assessment cycles. Each panel shows the assessed countries for that cycle (29 in 2023, 17 in 2024, 4 in 2025); the panels are not directly comparable as country samples differ.

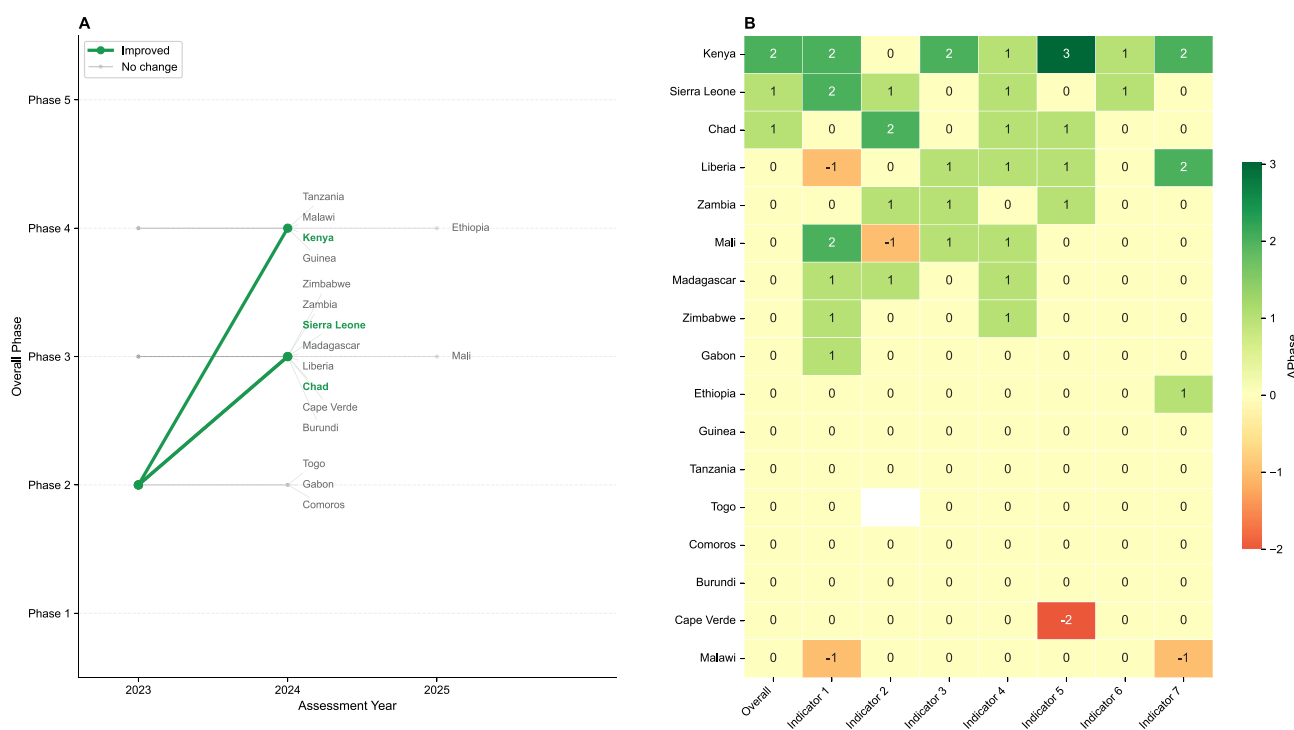


Figure 5 Slopegraph of overall phase trajectories and heatmap of overall and category-level phase change for countries with repeat assessments. **(A)** is a slope graph of overall phase between each country's earliest and latest assessment for the 17 repeat-assessed countries; the green-highlighted trajectories (Kenya, Sierra Leone, Chad) are the three countries that improved in overall phase. **(B)** is a heatmap of phase change for the same 17 countries across the overall phase and each of the 7 indicator categories. Indicators 1 to 7 denote leadership and governance, strategy and investment, legislation and policy, workforce, standards and interoperability, infrastructure, and services and applications, respectively.

The strongest correlation was between standards and interoperability and services and applications, with a coefficient of 0.70 ($p < 0.001$), followed by leadership and governance with workforce ($p = 0.59$, $p < 0.001$), and workforce with services and applications ($p = 0.59$, $p < 0.001$). The only notable negative association was a small, non-significant coefficient between strategy, investment, and infrastructure. The Kendall tau-b values were cross-checked for the same ordering of associations ([Supplementary S3](#)).

Hierarchical clustering with Ward's linkage identified three distinct country archetypes ([Figure 7](#)). Silhouette scores were 0.231, 0.193, and 0.200 for 2-, 3-, and 4-cluster Ward solutions, respectively; the gap statistic rose from

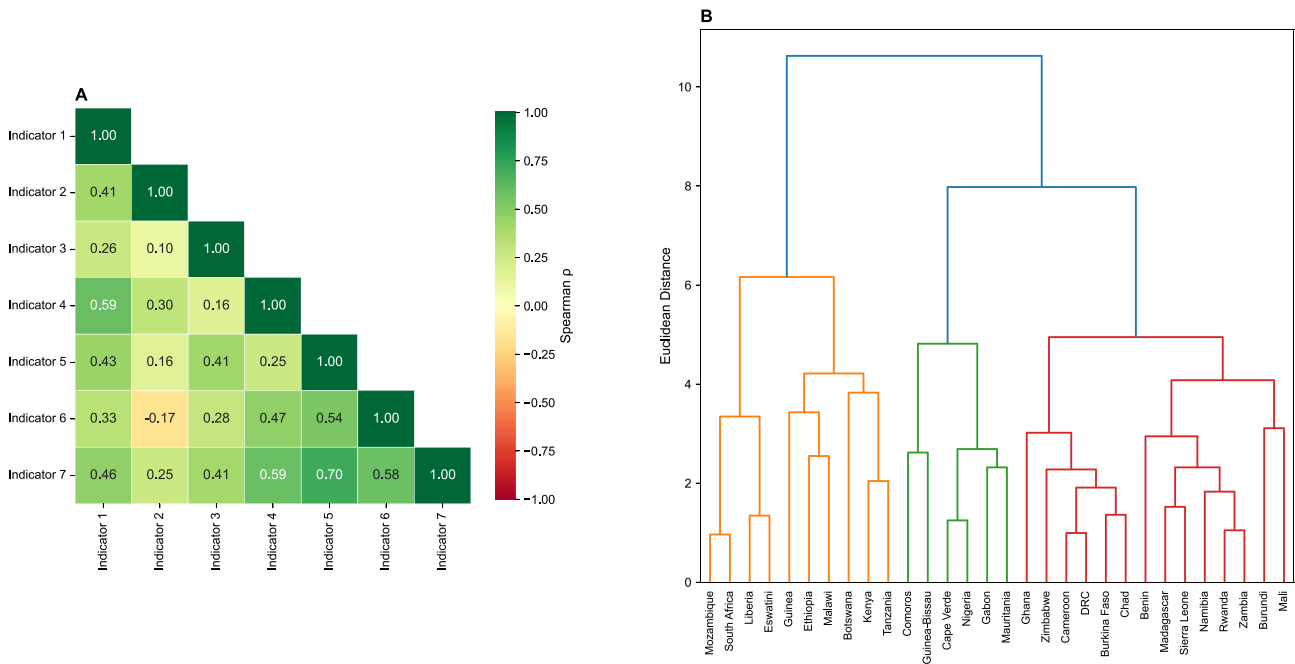


Figure 6 Spearman rank correlation matrix and hierarchical clustering dendrogram for the 7 GDHM category composite scores. **(A)** presents the Spearman correlation matrix using each country's latest assessment ($n = 30$ after listwise deletion). **(B)** displays the hierarchical clustering dendrogram (Ward linkage on z-standardized scores). Indicators 1 to 7 denote leadership and governance, strategy and investment, legislation and policy, workforce, standards and interoperability, infrastructure, and services and applications, respectively.

0.215 at $k = 2$ to a near-plateau of 0.28 at $k = 3$. Alternative linkage methods (single, complete, and average) produced one dominant cluster plus one to two small or singleton clusters. The mean domain scores by cluster are presented in Table 4. Cluster 1, comprising 10 countries (Botswana, Eswatini, Ethiopia, Guinea, Kenya, Liberia, Malawi, Mozambique, South Africa, and Tanzania), was characterized by relatively higher scores in standards and interoperability, infrastructure, and services and applications. Cluster 2, comprising 6 countries (Cape Verde, Comoros, Gabon, Guinea-Bissau, Mauritania, and Nigeria), exhibited the lowest scores across all categories, with particularly low workforce and services and applications. Cluster 3, comprising 14 countries (Benin, Burkina Faso, Burundi, Cameroon, Chad, the Democratic Republic of the Congo, Ghana, Madagascar, Mali, Namibia, Rwanda, Sierra Leone, Zambia, and Zimbabwe), showed relative strength in strategy, investment, leadership, and governance, but lower standards, interoperability, and workforce.

Within-Country Maturity Gaps

A stacked bar display of the phase distribution by category showed marked disparities across domains (Figure 8). Workforce had the largest combined share of countries in phase 1 or phase 2, at 68.8%, followed by standards and interoperability at 53.1%. By contrast, leadership and governance had no countries in phase 1 and the highest share in phase 4, reaching 31.2% of countries.

The within-country gap, defined as the difference between a country's maximum and minimum category phases, had a mean of 2.03 and a median of 2.00 across the 32 countries. Twenty-six of the 32 countries had gaps of two or more phases across their strongest and weakest domains. Seven countries had the maximum observed gap in the three phases: Mauritania, Burundi, Kenya, Tanzania, Niger, Mali, and Ghana. Six countries had a minimum gap of one phase. Among the countries with the largest gaps, the most common pattern was strong governance or legislation paired with a weak workforce in the health sector. Tanzania illustrated this pattern sharply, with phase 5 in leadership and governance and phase 2 in workforce, while Kenya showed phase 5 in standards and interoperability and phase 2 in workforce.



Figure 7 Radar chart profiles of the 3 country archetypes identified by hierarchical clustering on category composite scores. Cluster mean scores are also tabulated in Table 4 to allow numerical comparison alongside the radar visualization. Indicators 1 to 7 denote leadership and governance, strategy and investment, legislation and policy, workforce, standards and interoperability, infrastructure, and services and applications, respectively.

Discussion

This descriptive, multi-country analysis of 50 country-year observations from 32 African region member states provides one of the most comprehensive harmonized characterizations of digital health maturity across the WHO African Region published to date. Five findings are noteworthy in this study. First, most assessed countries are at an established level of overall maturity (phase 3), with no country at the nascent (phase 1) or optimized (phase 5) stages, indicating that the assessed countries have moved beyond the era of isolated pilots but have not yet achieved institutionalized and fully scaled digital health systems.² Second, a consistent cross-cutting pattern is evident across all three assessment cycles, in which leadership and governance are the strongest domains and workforce the weakest, pointing to a structural mismatch between institutional foundations and human capital readiness that is likely to constrain scale-up if left unaddressed. Third, among countries with repeat assessments, improvement outnumbered decline; Kenya's 2-phase advance within a single year is unusually large for a maturity-model construct and is consistent with the enactment of the Digital Health Act, 2023, and aligned investments, but should be interpreted with caution given the short observation window and potential changes in reporting completeness or government validation processes between cycles.^{18,25} Fourth, hierarchical clustering identified three distinct country archetypes corresponding to infrastructure-oriented, governance-led, and early stage profiles. Fifth, 26 of the 32 countries have maturity gaps of two or more phases across their strongest and weakest domains, providing an actionable typology for tailoring technical assistance and investment.

The prominence of leadership and governance is consistent with the momentum of the continental policy. Almost all African countries have published or are revising their national digital health strategies, many of which are aligned with the WHO Global Strategy on Digital Health and Africa CDC Digital Transformation Strategy. Recent analyses of African digital health strategic plans nonetheless caution that governance structures across the region remain uneven, with persistent gaps in workforce, legal frameworks, sustainable financing, and interoperability cited as recurring weaknesses

Table 4 Cluster Summary Statistics for the 3 Country Archetypes Identified by Hierarchical Clustering (Ward's Linkage, n = 30 Countries with Complete Category Data)

Cluster	N	L&G Mean Score	L&G Median Phase	S&I Mean Score	S&I Median Phase	Legis. Mean Score	Legis. Median Phase	Workforce Mean Score	Workforce Median Phase	Stds&Interop Mean Score	Stds&Interop Median Phase	Infra. Mean Score	Infra. Median Phase	Services Mean Score	Services Median Phase
1: Infrastructure-/ services-oriented	10	3.22	4.0	2.6	2.5	2.93	3.0	2.48	2.0	3.55	3.5	3.45	4.0	3.26	4.0
2: Early-stage	6	1.92	2.0	2.25	2.5	2.32	3.0	1.04	1.0	1.5	1.0	1.58	2.0	1.48	2.0
3: Governance-/ strategy-led	14	3.08	3.0	3.21	3.0	2.48	3.0	2.19	2.0	2.11	2.0	2.39	3.0	2.24	3.0

Notes: Clusters were derived by Ward's linkage hierarchical clustering on z-standardized category composite scores using each country's latest available assessment (30 countries with complete data across the 7 categories). Mean scores are the cluster mean of the category composite score; median phases are the cluster median of the ordinal category phase (1 to 5). Cluster numbering reflects the manuscript narrative (1 = infrastructure- and services-oriented; 2 = early-stage; 3 = governance- and strategy-led). Full cluster-validation outputs are reported in [Supplementary File S3](#).

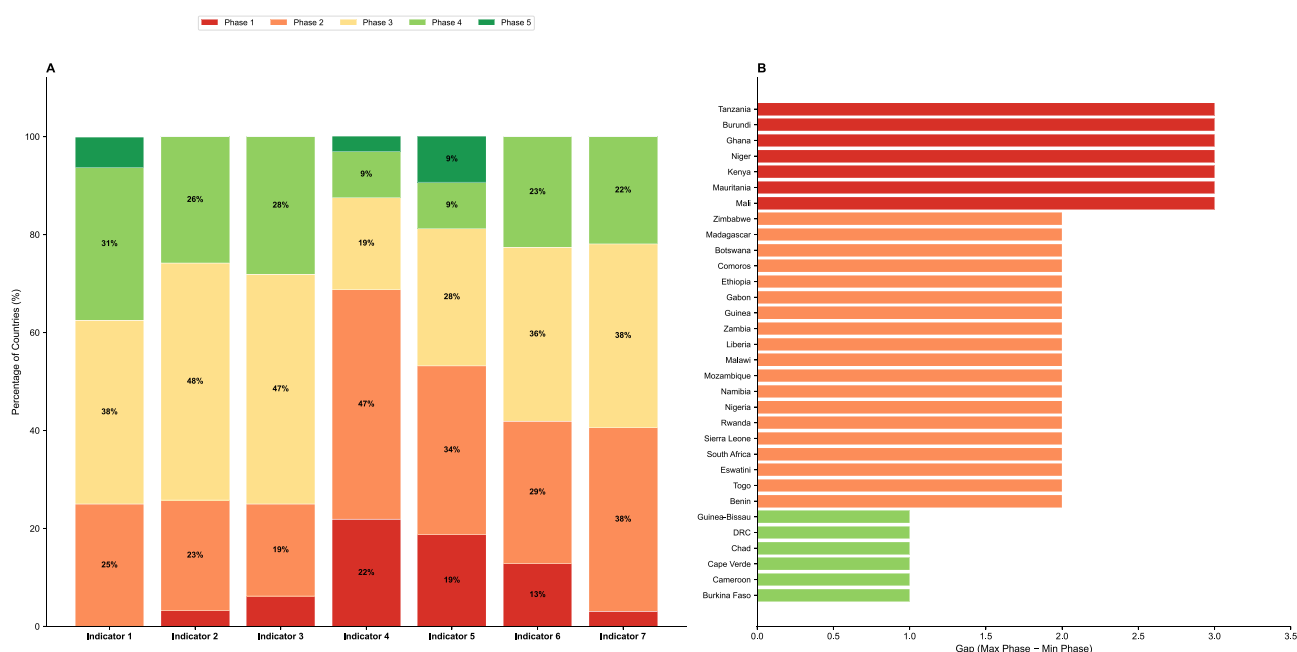


Figure 8 Phase distribution by category and within-country maturity gaps across the 32 assessed countries. **(A)** is a stacked bar chart showing the distribution of countries across the 5 phases for each of the 7 indicator categories. **(B)** is a horizontal bar chart of the within-country gap (maximum minus minimum category phase) for each country. Indicators 1 to 7 denote leadership and governance, strategy and investment, legislation and policy, workforce, standards and interoperability, infrastructure, and services and applications, respectively.

across reviewed plans.²⁶ These institutional foundations are necessary but insufficient. The positive association observed between standards and interoperability and services and applications is consistent with prior evidence that national investments in health data architecture co-occur with more mature services, although a cross-sectional correlation cannot establish the direction of effect; the association should therefore be read as a co-occurrence in cross-sectional maturity profiles rather than as a causal effect of investment in one domain on outcomes in another.²⁷ The persistent lag in workforce maturity mirrors long-standing concerns that digital health curricula are absent from most African pre-service health education programmes, that national health informatics cadres and career pathways remain underdeveloped, and that deployed systems frequently outpace the technical capacity to operate and sustain them.²⁸

The observation that short-term change among repeat-assessed countries is modest on average, yet punctuated by substantial gains in specific countries and domains, has two implications. First, aggregate regional progress should not be expected to follow a smooth trajectory over short horizons; measurable change occurs where concerted policy attention, financing, and implementation converge, as illustrated by Kenya's advances in standards, interoperability, legislation, and services between 2023 and 2024.²⁵ Second, domain-level gains often precede overall phase change because the overall phase aggregates across 7 categories and is insensitive to improvements in any single domain; domain-specific monitoring therefore provides earlier and more actionable signals than overall maturity alone.¹⁸

The country archetypes identified through hierarchical clustering extend previously proposed typologies of digital health readiness and suggest that a differentiated approach to technical cooperation is necessary. Concrete pathways for each archetype follow the cluster profiles. Infrastructure- and services-oriented countries (Cluster 1) would benefit most from sustained investment in the digital health workforce and translating their interoperability and service foundations into routinely measured improvements in care quality and equity. Governance- and strategy-led countries (Cluster 3) have the institutional scaffolding to scale, and should prioritize the operational steps that move strategies into deployed services: funded health information exchange, terminology and identity standards, and facility-level systems linked to performance indicators. Early-stage countries (Cluster 2) would gain most from foundational investment in leadership, legislation, baseline infrastructure, and workforce, sequenced to enable later interoperability and service investments.^{10,16,27}

Our findings sit alongside and complement the small but growing empirical literature on digital health in Africa. Country-level case studies from Rwanda, Kenya, South Africa, and Ethiopia have documented the adoption of electronic medical records, integrated disease surveillance, and national health information exchanges but have typically assessed single technologies or settings rather than the broader enabling environment.¹⁷ Regional policy reviews have noted the rapid expansion of national digital health strategies but have highlighted a gap in comparable outcome measurements.⁹ By analyzing 50 country-year observations through a harmonized framework, this study provides regional evidence on where the assessed Member States of the WHO African Region stand collectively and how domains relate to one another, contributing to the regional baseline that has been called for in the global digital health monitoring agenda.²

Comparison with high-income digital health systems should be made with caution: recent Organization for Economic Co-operation and Development (OECD) reporting shows that advanced economies operate at maturity levels not captured by the 5-phase GDHM rubric used here, with countries such as Denmark, Estonia, and several other European systems offering national patient portals with comprehensive electronic health data and unique person identifiers for cross-domain linkage.²⁹ The absence of any African country in phase 5 in our analysis, therefore, reflects, in part, structural and historical differences in digital infrastructure investment and is not, on its own, evidence of a uniform readiness deficit.

This study had several limitations that should be considered when interpreting the findings. Data were self-reported by country partners and ministries of health, and government approval is required for publication, which may introduce social desirability and selection biases, potentially inflating maturity in high-performing domains and suppressing less favorable findings. Only 32 of the 47 African region Member States of the WHO had published assessments over the study period, and participation was not random; the sample may skew towards countries with active digital health partnerships, ongoing national monitoring capacity, or existing reform agendas, which limits external generalizability to all 47 member states and to the wider African continent. Assessment coverage declined over the study period, from 29 countries in 2023 to four in 2025; therefore, year-specific comparisons of regional means must be interpreted cautiously, the 2025 subset is too small to support trend inference, and the small number of repeat assessments precluded inferential trend modelling.

The narrow 2023–2025 reporting window and the unbalanced panel mean that our analysis is best treated as a short-term descriptive baseline rather than a longitudinal study of maturity trajectories. The GDHM measures the enabling environment rather than the clinical or population health outcomes that ultimately justify digital health investments. Further research is needed to link maturity scores to service delivery, quality, equity, and health outcomes and to examine how structural factors, such as gross domestic product per capita, broader information and communications technology infrastructure, and educational attainment, shape national digital health rankings. Finally, composite scores are aggregated across heterogeneous indicators, and small differences in category scores should be interpreted as summary maturity signals rather than exact performance estimates.

Despite these limitations, the findings of this study have implications for policies, investments, and research. African countries, with support from the African Union, regional economic communities, the WHO Regional Office for Africa, and development partners, should prioritize sustained investment in the digital health workforce, including the integration of digital and data science competencies into pre-service curricula, the creation of formal career structures for health informaticians, and in-service training linked to deployed systems.²⁸ Investment in interoperability standards and shared health data architectures appears to be associated with more mature services and applications, suggesting potential value, rather than a guaranteed return, from continued work on national digital health blueprints, health information exchanges, and harmonized data standards across the region.²⁷ Tailored technical cooperation informed by country archetypes, coordinated regional investment in digital public infrastructure, and explicit integration of AI governance and cybersecurity into national digital health strategies are likely to be the next frontiers of regional policy.³⁰ Finally, expanding the country coverage and annual frequency of the GDHM and linking its maturity indicators to service delivery and health outcome data would substantially strengthen the evidence base for monitoring progress towards the WHO Global Strategy on Digital Health and continental digital transformation commitments.

Conclusion

Digital health maturity among the assessed African region member states has reached an established baseline in most cases, with leadership and governance emerging as the strongest domain and the workforce as the weakest, and standards and interoperability unevenly distributed across the region. Geographic patterns were heterogeneous: phase-4 maturity was distributed across East, West, and Southern Africa rather than concentrated in a single sub-region, while phase-2 maturity clustered in parts of West and Central Africa, indicating that location alone is a poor predictor of digital health maturity and that policy attention and partnership ecosystems likely matter more than geography. Short-term descriptive change among the 17 repeat-assessed countries pointed to modest improvement on average, with Kenya advancing two overall phases in a single year; this should be read as suggestive of what concerted policy attention can achieve, not as evidence of region-wide acceleration. Clustering revealed three distinct country archetypes corresponding to infrastructure-oriented, governance-led, and early stage profiles, which can support tailored technical assistance and investment. Within-country gaps were common and most often reflected strong governance performance paired with weak workforce capacity, indicating a persistent mismatch between institutional foundations and human capital readiness. Strengthening the digital health workforce, investing in interoperability and shared data architectures, and expanding participation in structured maturity assessments are likely to be the most consequential levers for advancing digital health in the WHO African Region. These findings apply to the 32 Member States with published GDHM data from 2023 to 2025 and should not be extrapolated to all 47 Member States of the WHO African Region or the wider African continent. Broader country participation, repeated annual measurements, independent validations, and linkages with service delivery and health outcome indicators are needed to support stronger generalizations and trend inferences in future cycles.

Disclosure of AI Use

The authors used Paperpal AI (<https://paperpal.com>) for language editing to improve the clarity and readability of the manuscript. This assistance was limited to linguistic refinement only; the study design, data analysis, interpretation, and all conclusions remain the responsibility of the authors.

Data Sharing Statement

Aggregated country-level digital health maturity data were accessed from the Global Digital Health Monitor (GDHM), health.enabled, World Health Organization and partners. Available at: <https://digitalhealthmonitor.org>. Date accessed: April 2026 (2023–2025 cycles).

Ethics Statement

This study did not involve any human data. It analyzed publicly available, aggregated, country-level digital health maturity indicators from the Global Digital Health Monitor and did not involve identifiable individual subjects. The Institutional Review Board of SIMAD University reviewed the protocol and confirmed that the study was exempt from a full ethics review on the basis that it does not constitute human subjects research (exemption reference 2026/SU-IRB/P00194).

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Disclosure

The authors declare no conflicts of interest in this work.

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